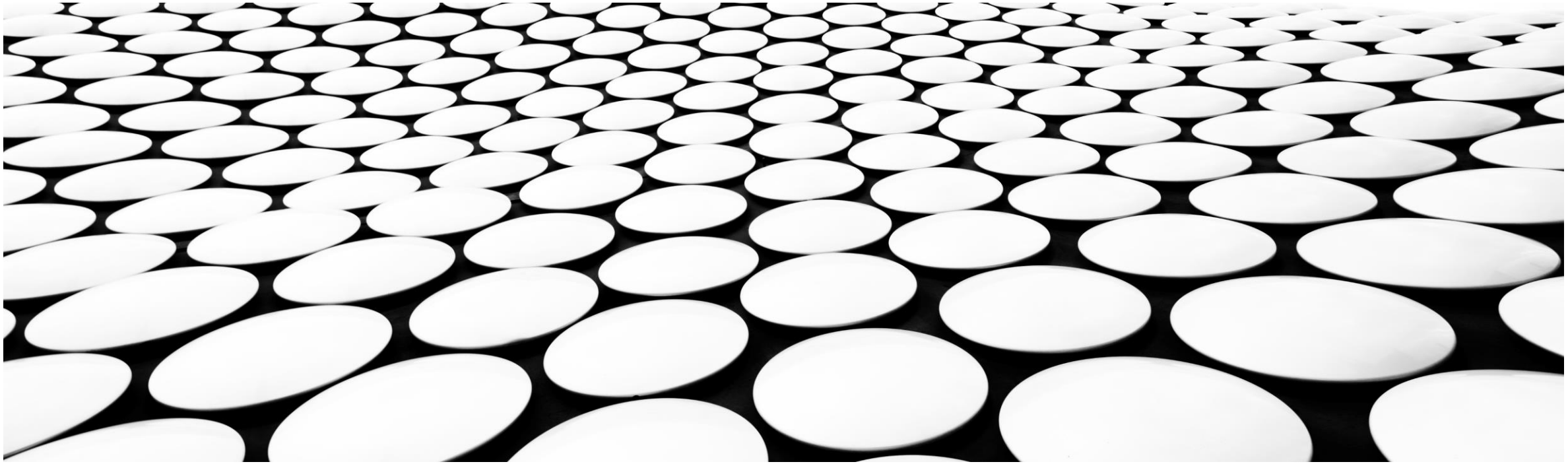


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Endpoint Security

Engineering Cybersecurity Solutions

Donnacha Forde



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$whoami

- Developer
- Architect
- Manager
- Technical Director
- Technology Strategist



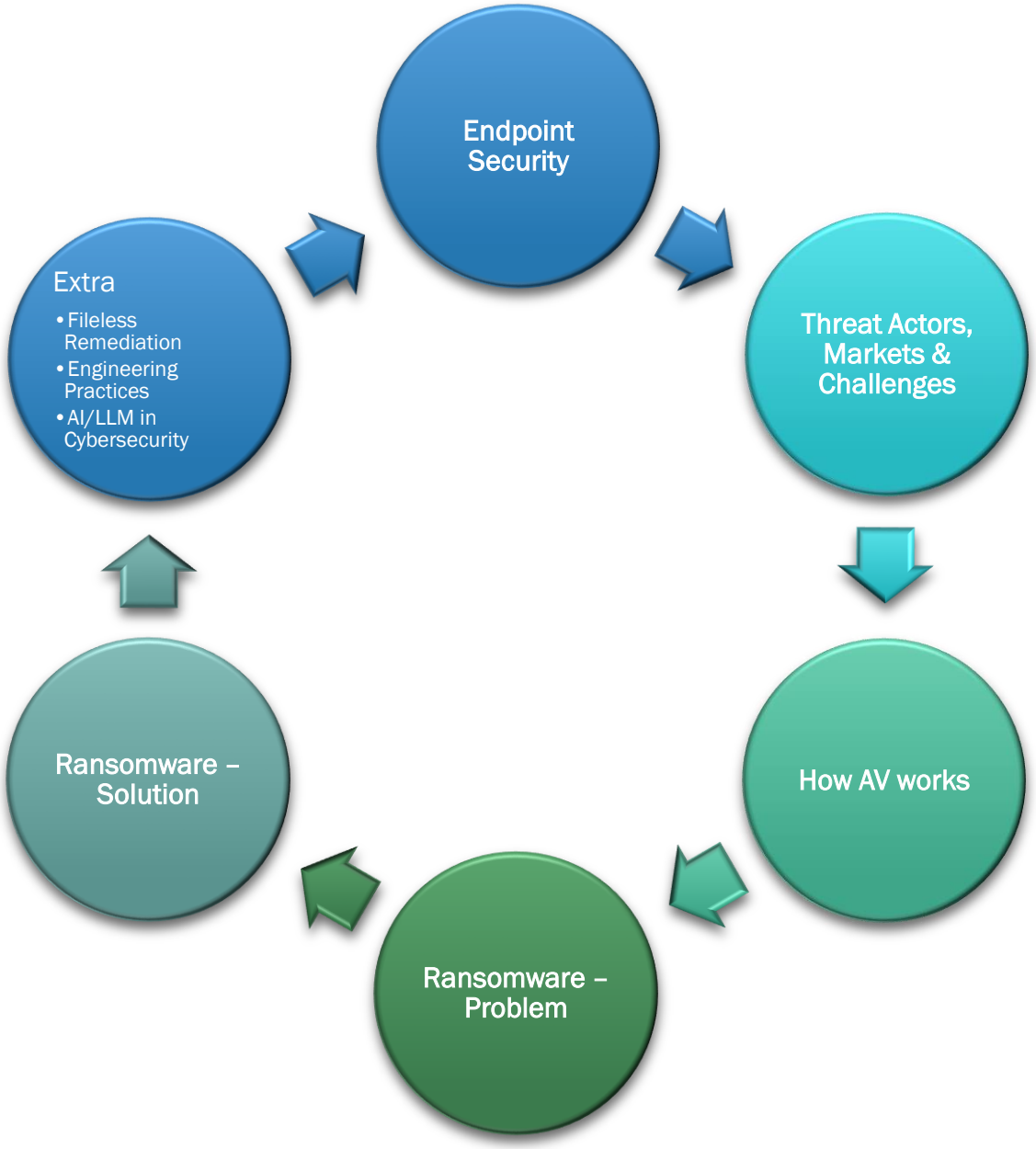
Mostly Finance (~10yrs)

Mostly Systems-Management (~15yrs)

Security (~10 yrs)

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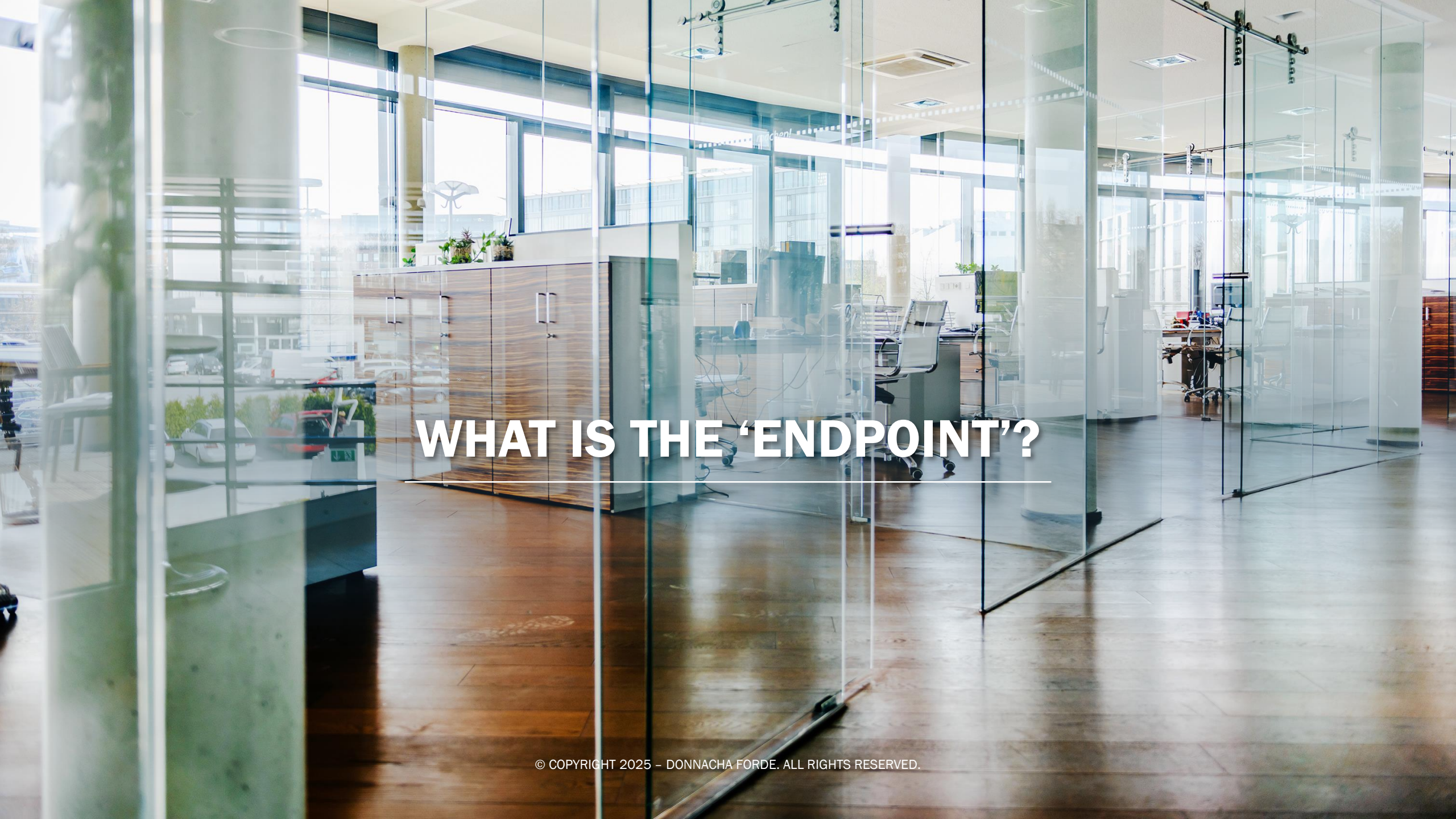
AGENDA



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Endpoint Security

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WHAT IS THE ‘ENDPOINT’?

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THE ENDPOINT



Desktop PC



Laptop



Tablet



Mobile



Server



Gaming Device



IoT Device



Smart Home Devices



VMs & Containers



Network

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ECOSYSTEMS



Windows



macOS



Linux



ChromeOS



iOS



Android



x64



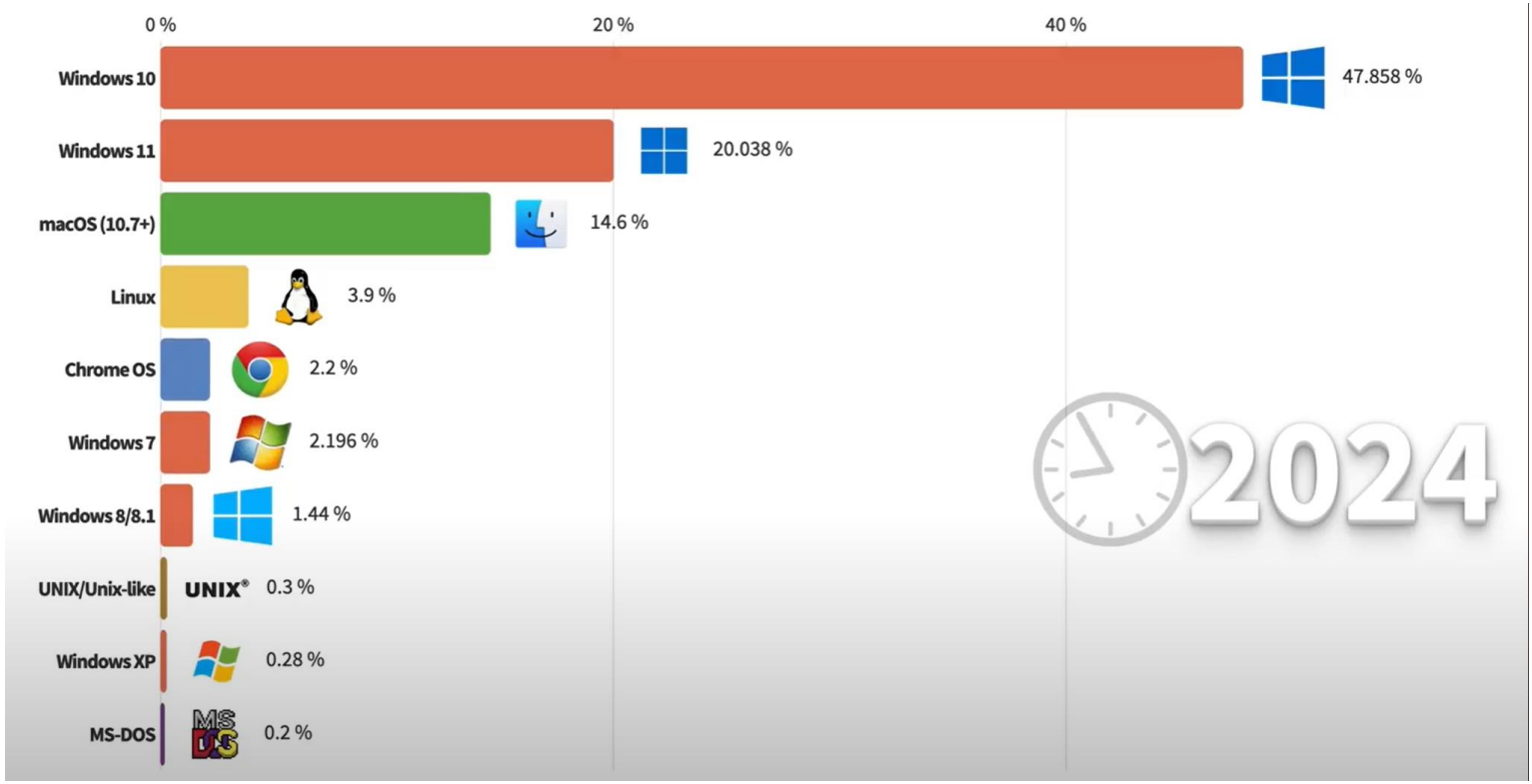
arm64

Reference: https://donnachaforde.github.io/blogs+posts/software-engineering/wouldnt-start-here/

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OS Prevalence – Market Share

- Windows 10 & 11 >67%
- macOS ~15%
- Linux ~4%
- ChromeOS ~2%



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Source: https://www.youtube.com/watch?v=Tjc-qrjzRcU

WHAT IS ENDPOINT SECURITY?



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ENDPOINT SECURITY

Antivirus (AV) / Antimalware	Data Loss Prevention (DLP) / Data Protection	Ad Blocking
Patch & Vulnerability Management	Password Management	Email protection*
Identity Protection* (IAM)	Endpoint Detection & Response (EDR)	Web Protection
Browser Isolation	Firewall	Network Control/Protection
Intrusion Detection and Prevention (IDP)	VPN	DNS Filtering



Core Endpoint Technologies

EPP – Antivirus and Antimalware

- Detection
  - Signature-based Detection → Heuristics → Rule-based → Behaviour-based
- On demand scanning/On access scanning
- Real-time Protection
  - Monitor Files/Processes, Downloads and Network Traffic
- Quarantine
- Remediation
- Anti-ransomware
- Internal
  - Self-Protection/Tamper Detection
  - Trust → Performance

Endpoint Detection & Response

- Real-time Monitoring
- Visibility
- Detection
  - Automatic via Content Rules
  - Manually by SOC Analysts
- Analysis
  - Historic Queries
  - Distributed Queries
- Response
  - Remote Access
  - Network Isolation
- Continuous Data Collection → SOC Analysts/Threat Hunting

Reference: https://donnachaforde.github.io/blogs+posts/software-engineering/endpoint-protection/

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EPP versus EDR

Natural Tension:

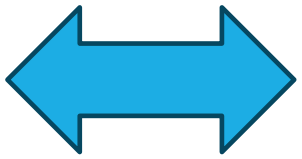
Protection v Detection



EPP

Prevention

Blocking





EDR

Monitoring

Detection

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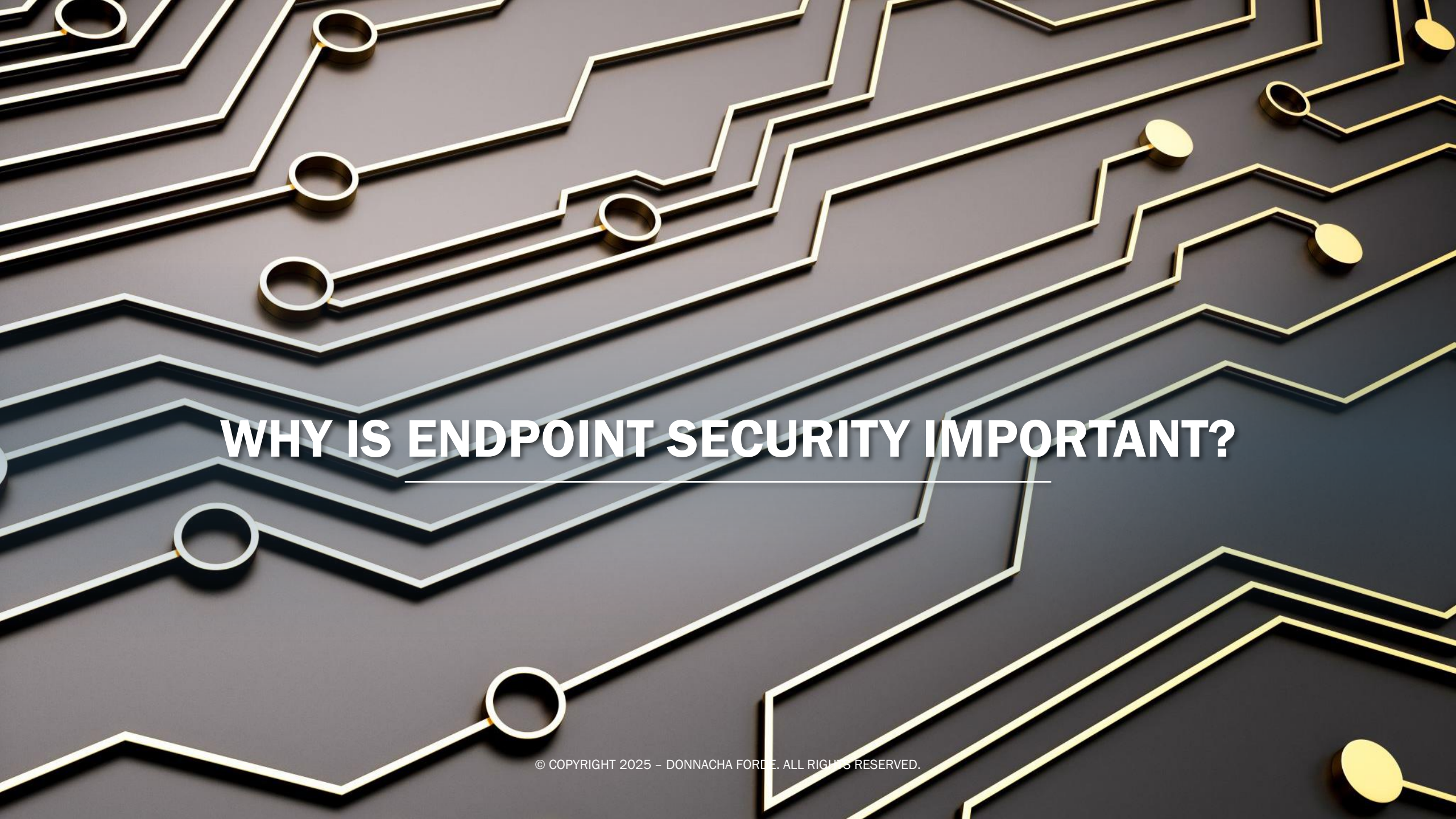
Cloud Infrastructure

- Systems Management
  - Multi-tenant Cloud
  - Managed Solutions (e.g. MDR)
  - Large no. of users
  - Large no. of endpoints
  - Huge volume of events

- Endpoint Support System
  - Client/Server solutions
  - Cloud-based Solutions
    - e.g. ML Models, Behavioural Analysis, Detonation/Sandbox
  - Event Management
    - Messaging
    - Queue Management
    - Persistence (e.g. Graphs)
  - IPC
    - Distributed Queries (Cloud → Endpoint)
    - Lookup (Endpoint → Cloud)
  - Performance ←→ POP



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WHY IS ENDPOINT SECURITY IMPORTANT?

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WHY IS ENDPOINT SECURITY STILL SO CRITICAL?



Ever increasing number of connected devices



Industry trends

Bring your own device (BYOD)

Shadow IT

Remote Working (WFH)

Shift to Cloud Computing



Prevalence of Real-world threats

Ransomware

Data breaches

Insider-threats



Cost of Security Incidents

Direct extorsion - ransom

Financial loss – impact on business function

Reputational damage

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COSTS OF CYBERCRIME

Estimated cost globally

- 2021 → > **$6 Trillion**
- 2022 → > **$8 Trillion**
- 2023 → > **$11 Trillion**
- 2026 → $20 Trillion*

World's 3rd largest economy (after USA and China)

Doubled since 2015

Almost doubled again since 2021

Context

- More profitable than Global Trade in Illegal Drugs
- Costs are greater costs than the damage from natural disasters
- Cybercrime represents the greatest transfer of economic wealth in history

Source: https://www.evolvesecurity.com/blog-posts/actual-cost-of-cybercrime

Source: https://cybersecurityventures.com/hackerpocalypse-cybercrime-report-2016

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*estimated

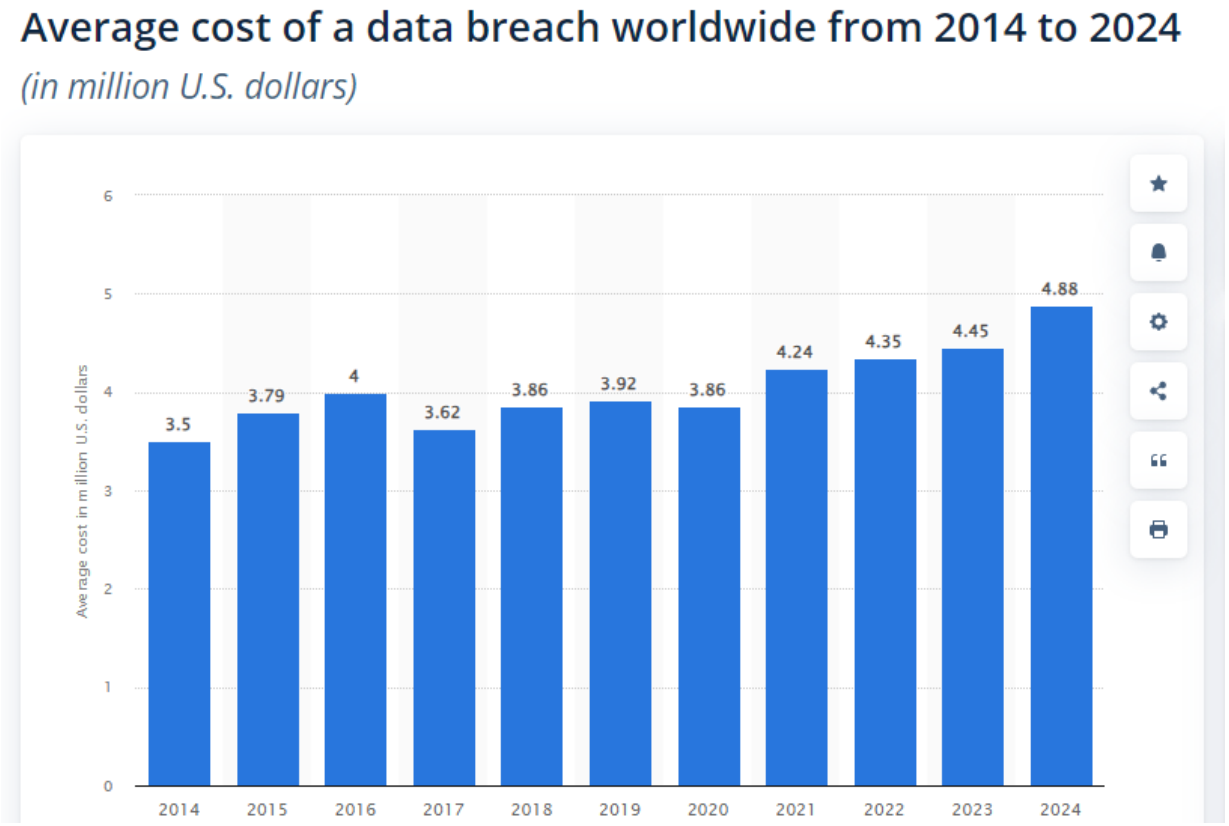


Costs of Data Breaches

- Data Breach Costs
  - Average cost of data breach – all sectors: $4.88M
    - (10% increase over 2023)
  - Average cost of data breach in the financial sector: $6.08M
    - (3% increase over 2023)

- Estimated 70% of successful data breaches **originate at the endpoint**

Source: https://www.evolvesecurity.com/blog-posts/actual-cost-of-cybercrime

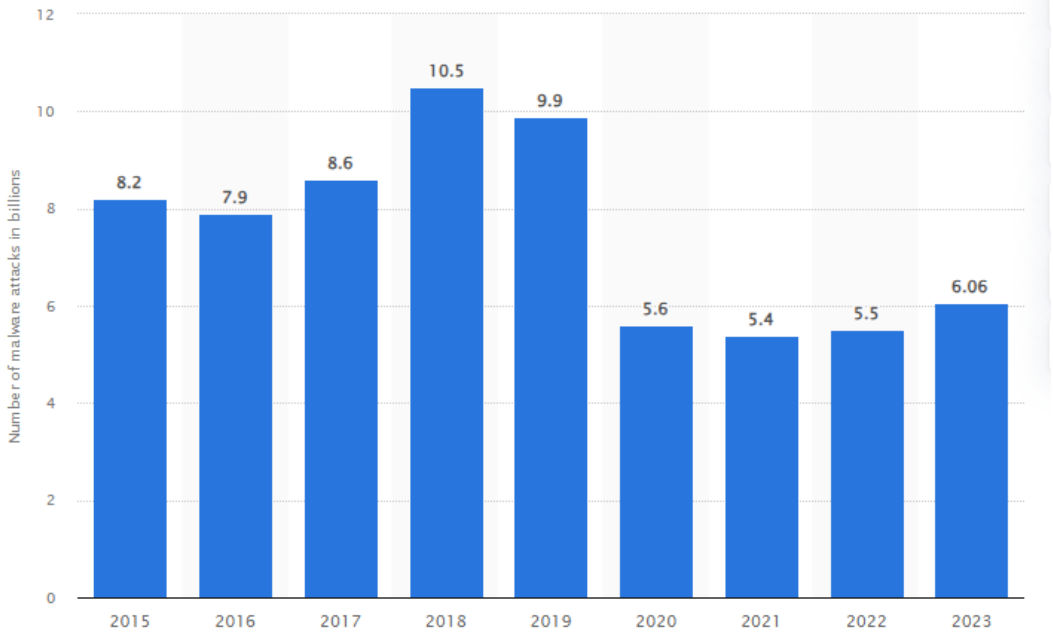


Source: https://www.statista.com/statistics/987474/global-average-cost-data-breach

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Annual Number of Malware Attacks

- Period: 2015 – 2023
- 6.06 billion in 2023
- 10% increase on previous year
- 10.5 billion attacks in 2018
- Websites are the most common attack vector



Source: https://www.statista.com/statistics/873097/malware-attacks-per-year-worldwide/

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Adversaries, Market & Challenges

Who are the threat actors? Who are we protecting and what are the challenges?

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Where are the threats coming from?

Top three threats



Organized Crime

Criminal gangs

Often state-sponsored and/or state sanctioned



Nation-state

Political and/or Financial motives



Hacktivism

Disruption

Exposure

Access Data

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Famous Chollima



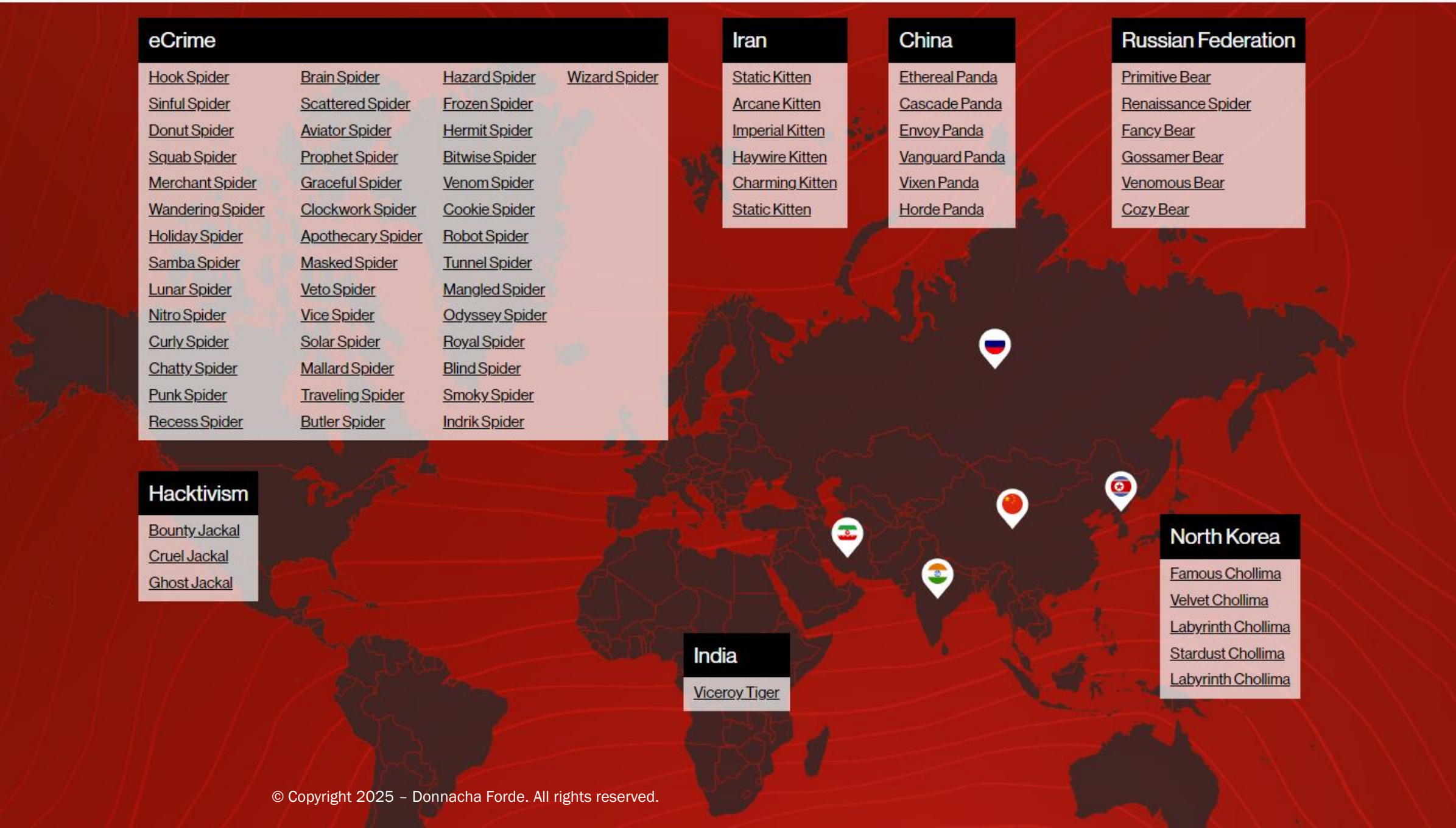
Punk Spider



Horde Panda

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Global Threat Landscape



eCrime

Hook Spider	Brain Spider	Hazard Spider	Wizard Spider
Sinful Spider	Scattered Spider	Frozen Spider	
Donut Spider	Aviator Spider	Hermit Spider	
Squab Spider	Prophet Spider	Bitwise Spider	
Merchant Spider	Graceful Spider	Venom Spider	
Wandering Spider	Clockwork Spider	Cookie Spider	
Holiday Spider	Apothecary Spider	Robot Spider	
Samba Spider	Masked Spider	Tunnel Spider	
Lunar Spider	Veto Spider	Mangled Spider	
Nitro Spider	Vice Spider	Odyssey Spider	
Curly Spider	Solar Spider	Royal Spider	
Chatty Spider	Mallard Spider	Blind Spider	
Punk Spider	Traveling Spider	Smoky Spider	
Recess Spider	Butler Spider	Indrik Spider	

Iran

- Static Kitten
- Arcane Kitten
- Imperial Kitten
- Haywire Kitten
- Charming Kitten
- Static Kitten

China

- Ethereal Panda
- Cascade Panda
- Envoy Panda
- Vanguard Panda
- Vixen Panda
- Horde Panda

Russian Federation

- Primitive Bear
- Renaissance Spider
- Fancy Bear
- Gossamer Bear
- Venomous Bear
- Cozy Bear

Hacktivism

- Bounty Jackal
- Cruel Jackal
- Ghost Jackal

India

- Viceroy Tiger

North Korea

- Famous Chollima
- Velvet Chollima
- Labyrinth Chollima
- Stardust Chollima
- Labyrinth Chollima

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ATT&CKcon 6.0 returns October 14-15, 2025 in McLean, VA. More details about tickets and our CFP can be found here

ATT&CK®

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Contribute | Blog

FAQ | Random Page

MITRE ATT&CK® is a globally-accessible knowledge base of adversary tactics and techniques based on real-world observations. The ATT&CK knowledge base is used as a foundation for the development of specific threat models and methodologies in the private sector, in government, and in the cybersecurity product and service community.

With the creation of ATT&CK, MITRE is fulfilling its mission to solve problems for a safer world — by bringing communities together to develop more effective cybersecurity. ATT&CK is open and available to any person or organization for use at no charge.

ATT&CK Matrix for Enterprise

layout: side | show sub-techniques | hide sub-techniques

Reconnaissance	Resource Development	Initial Access	Execution	Persistence	Privilege Escalation	Defense Evasion	Credential Access	Discovery	Lateral Movement	Collection	Command and Control	Exfiltration	Impact
10 techniques	8 techniques	10 techniques	14 techniques	20 techniques	14 techniques	44 techniques	17 techniques	32 techniques	9 techniques	17 techniques	18 techniques	9 techniques	14 techniques
Active Scanning (3)	Acquire Access	Content Injection	Cloud Administration Command	Account Manipulation (7)	Abuse Elevation Control Mechanism (6)	Abuse Elevation Control Mechanism (6)	Adversary-in-the-Middle (4)	Account Discovery (4)	Exploitation of Remote Services	Adversary-in-the-Middle (4)	Application Layer Protocol (5)	Automated Exfiltration (1)	Account Access Removal
Gather Victim Host Information (4)	Acquire Infrastructure (8)	Drive-by Compromise	Command and Scripting Interpreter (11)	BITS Jobs	Access Token Manipulation (5)	Access Token Manipulation (5)	Brute Force (4)	Application Window Discovery	Internal Spearphishing	Archive Collected Data (3)	Communication Through Removable Media	Data Transfer Size Limits	Data Destruction (1)
Gather Victim Identity Information (3)	Compromise Accounts (3)	Exploit Public-Facing Application	Container Administration Command	Boot or Logon Autostart Execution (14)	Account Manipulation (7)	BITS Jobs	Credentials from Password Stores (6)	Browser Information Discovery	Lateral Tool Transfer	Audio Capture	Content Injection	Exfiltration Over Alternative Protocol (3)	Data Encrypted for Impact
Gather Victim Network Information (6)	Compromise Infrastructure (8)	External Remote Services	Deploy Container	Boot or Logon Initialization Scripts (5)	Boot or Logon Autostart Execution (14)	Build Image on Host	Exploitation for Credential Access	Cloud Infrastructure Discovery	Remote Service Session Hijacking (2)	Automated Collection	Data Encoding (2)	Exfiltration Over C2 Channel	Data Manipulation (3)
Gather Victim Org Information (4)	Develop Capabilities (4)	Hardware Additions	Exploitation for Client Execution	Browser Extensions	Boot or Logon Initialization Scripts (5)	Debugger Evasion	Forced Authentication	Cloud Service Dashboard	Remote Services (8)	Browser Session Hijacking	Data Obfuscation (3)	Exfiltration Over Other Network Medium (1)	Defacement (2)
Phishing for Information (4)	Establish Accounts (3)	Phishing (4)	Inter-Process Communication (3)	Compromise Host Software Binary	Create or Modify System Process (5)	Deobfuscate/Decode Files or Information	Forge Web Credentials (2)	Cloud Service Discovery	Replication Through Removable Media	Clipboard Data	Dynamic Resolution (3)	Exfiltration Over Physical Medium (1)	Disk Wipe (2)
Search Closed Sources (2)	Obtain Capabilities (7)	Replication Through Removable Media	Native API	Create Account (3)	Domain or Tenant Policy Modification (2)	Deploy Container	Input Capture (4)	Cloud Storage Object Discovery	Software Deployment Tools	Data from Cloud Storage	Encrypted Channel (2)	Exfiltration Over Web Service (4)	Endpoint Denial of Service (4)
Search Open Technical Databases (5)	Stage Capabilities (6)	Supply Chain Compromise (3)	Scheduled Task/Job (5)	Create or Modify System Process (5)	Escape to Host	Direct Volume Access	Modify Authentication Process (9)	Container and Resource Discovery	Taint Shared Content	Data from Configuration Repository (2)	Fallback Channels	Scheduled Transfer	Financial Theft
Search Open Websites/Domains (3)		Trusted Relationship	Serverless Execution	Event Triggered Execution (17)	Event Triggered Execution (17)	Domain or Tenant Policy Modification (2)	Multi-Factor Authentication Interception	Debugger Evasion	Use Alternate Authentication Material (4)	Data from Information Repositories (5)	Hide Infrastructure	Transfer Data to Cloud Account	Firmware Corruption
Search Victim-Owned Websites		Valid Accounts (4)	Shared Modules	External Remote Services	Exploitation for Privilege Escalation	Execution Guardrails (2)	Multi-Factor Authentication Request Generation	Device Driver Discovery		Data from Local System	Ingress Tool Transfer		Inhibit System Recovery
			Software Deployment Tools	Hijack Execution Flow (13)	Hijack Execution Flow (13)	Exploitation for Defense Evasion	Network Sniffing	Domain Trust Discovery		Data from Network Shared Drive	Multi-Stage Channels		Network Denial of Service (2)
			System Services (2)	Implant Internal Image	Process Injection (12)	File and Directory Permissions Modification (2)	OS Credential Dumping (8)	File and Directory Discovery		Data from Removable Media	Non-Application Layer Protocol		Resource Hijacking (4)
			User Execution (3)	Modify Authentication Process (9)	Scheduled Task/Job (5)	Hide Artifacts (12)	Steal Application Access Token	Group Policy Discovery		Data Staged (2)	Non-Standard Port		Service Stop
			Windows Management Instrumentation	Office Application Startup (6)	Valid Accounts (4)	Hijack Execution Flow (13)	Steal or Forge Authentication Certificates	Log Enumeration		Email Collection (3)	Protocol Tunneling		System Shutdown/Reboot
				Power Settings		Impair Defenses (11)	Steal or Forge Kerberos Tickets (5)	Network Service Discovery		Input Capture (4)	Proxy (4)		
				Pre-OS Boot (5)		Impersonation	Steal Web Session Cookie	Network Share Discovery		Screen Capture	Remote Access Software		
				Scheduled Task/Job (5)		Indicator Removal (10)	Unsecured Credentials (8)	Network Sniffing		Video Capture	Traffic Signaling (2)		
				Server Software Component (5)		Indirect Command Execution		Password Policy Discovery			Web Service (3)		
				Traffic Signaling (2)		Masquerading (10)		Peripheral Device Discovery					
				Valid Accounts (4)		Modify Authentication Process (9)		Permission Groups Discovery (3)					
						Modify Cloud Compute Infrastructure (5)		Process Discovery					
						Modify Cloud Resource Hierarchy		Query Registry					
						Modify Registry		Remote System Discovery					
						Modify System Image (2)		Software Discovery (1)					
						Network Boundary Bridging		System Information Discovery					
								System Location Discovery (1)					
								System Network Configuration Discovery (2)					

Market Segments



Enterprise

Primary Target for Adversaries

Environment

- Exist behind Firewall
- Network traffic goes through a gateway

Managed Endpoint

- More sterile environment, Less deviation



SME

Enterprise v SOHO



Consumer

More diverse environment

Not sterile

Greater exposure

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Endpoint Security Challenges

- Evolving Threats
  - Zero-day Attacks
  - Advanced Persistent Threats (APT)
  - Insider Threats
- Trends
  - Shadow IT
  - Remote Working
  - BYOD
- Complexity – Managing at Scale
  - Security & IT Management
- Reaction Time vs Volume of Data
  - SOC Analyst Response
  - Threat Hunting
  - Threat Intel



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Endpoint Engineering Challenges

- Scale
  - Volume of System Event Info
- Costs → COGS
  - CPU, Network, Storage
- Security vs User Convenience vs Performance
- Complex, large-scale environments
- Rapid Response
- Prompt Delivery of new features & solutions in a complex environment
- Risk & Impact



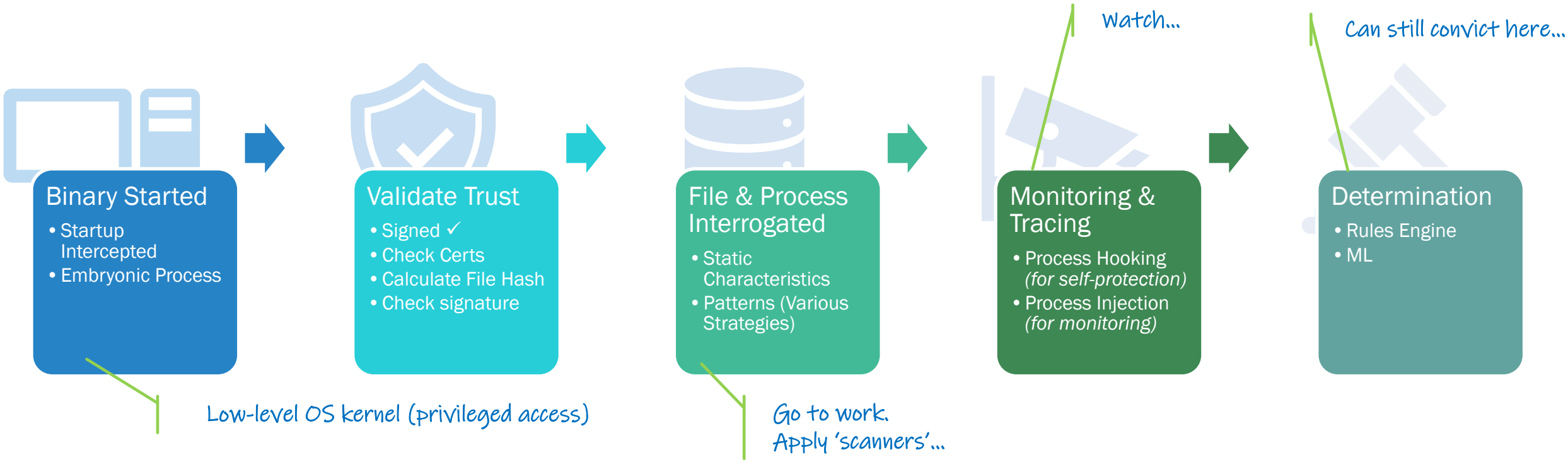
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Antivirus, Antimalware & EDR

How AV & Ransomware works

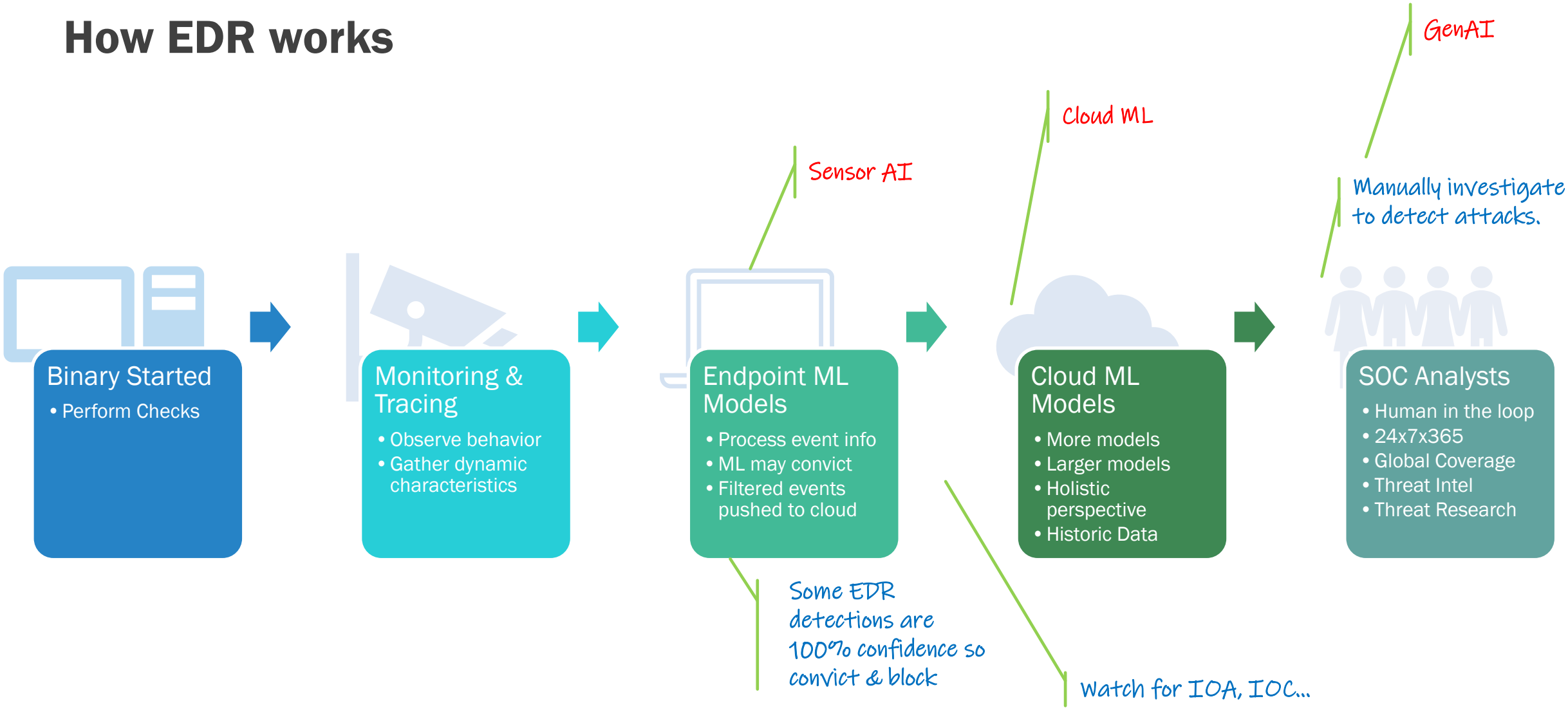
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How Traditional AV works



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How EDR works



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The Problem – Ransomware

A look at Ransomware Trends

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Ransomware continues to be a Threat

- US is the most attacked country.

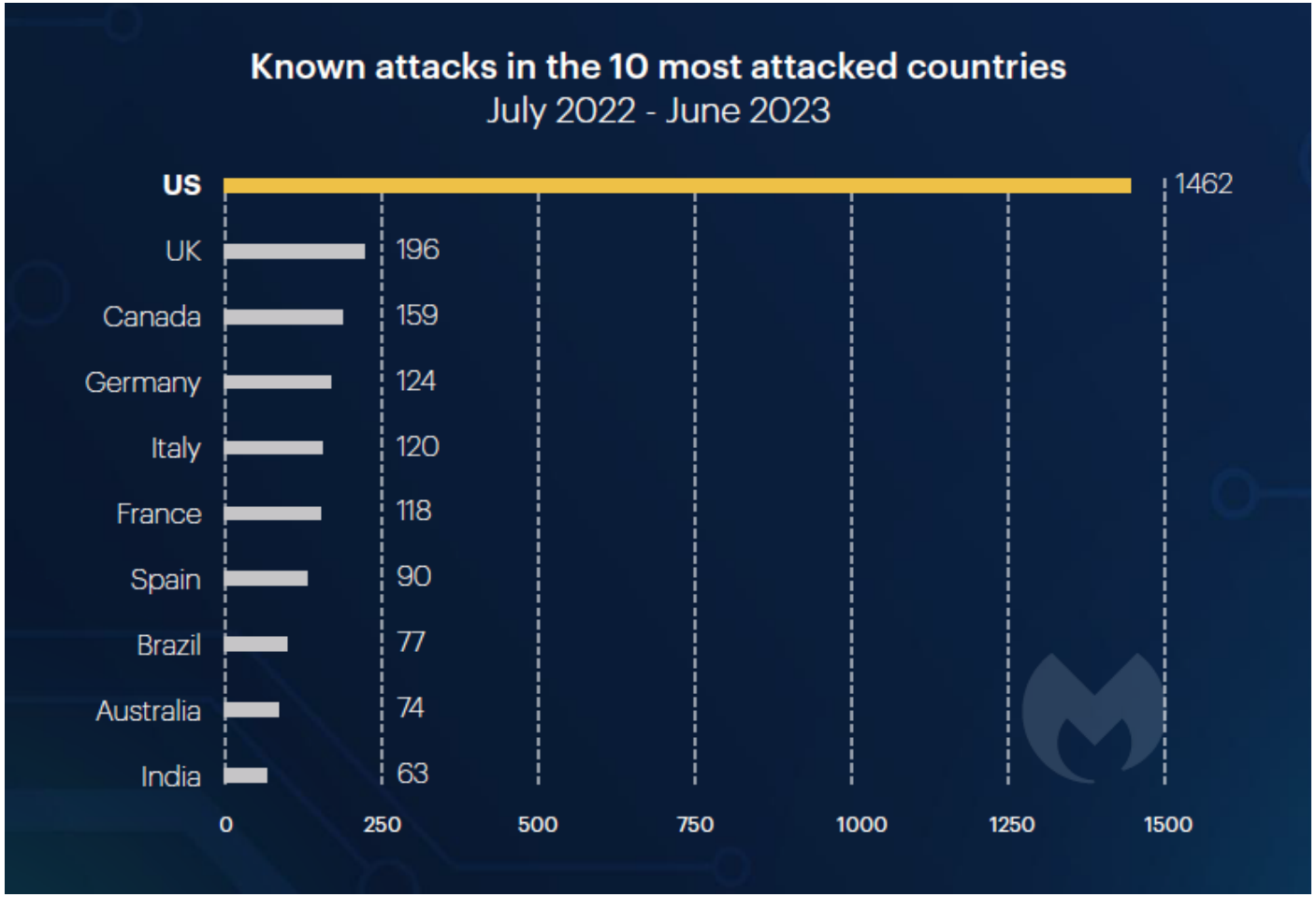


Image Credit: *Malwarebytes – State of Ransomware Report 2023*

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Ransomware continues to be a Threat

- Ransomware attacks in last 12 months.

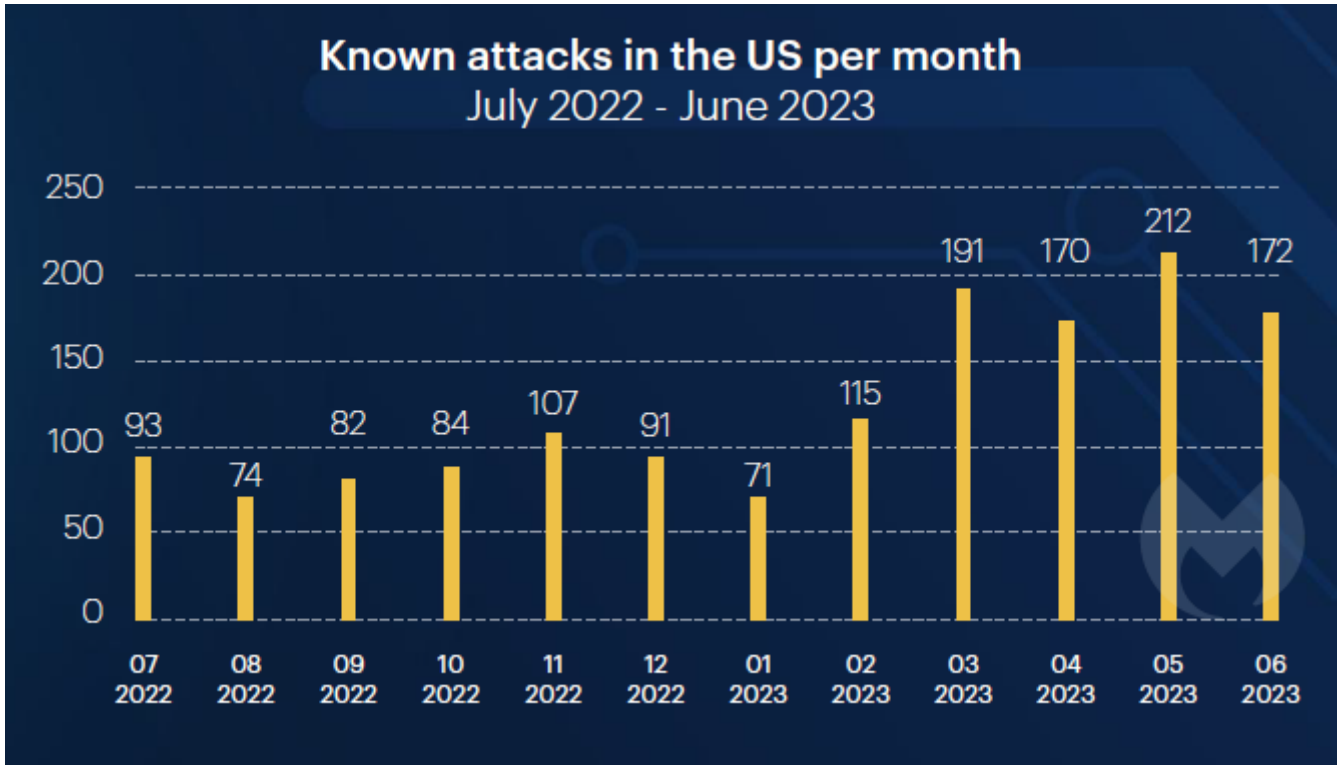


Image Credit: *Malwarebytes – State of Ransomware Report 2023*

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Ransomware continues to be a Threat

- Active gangs

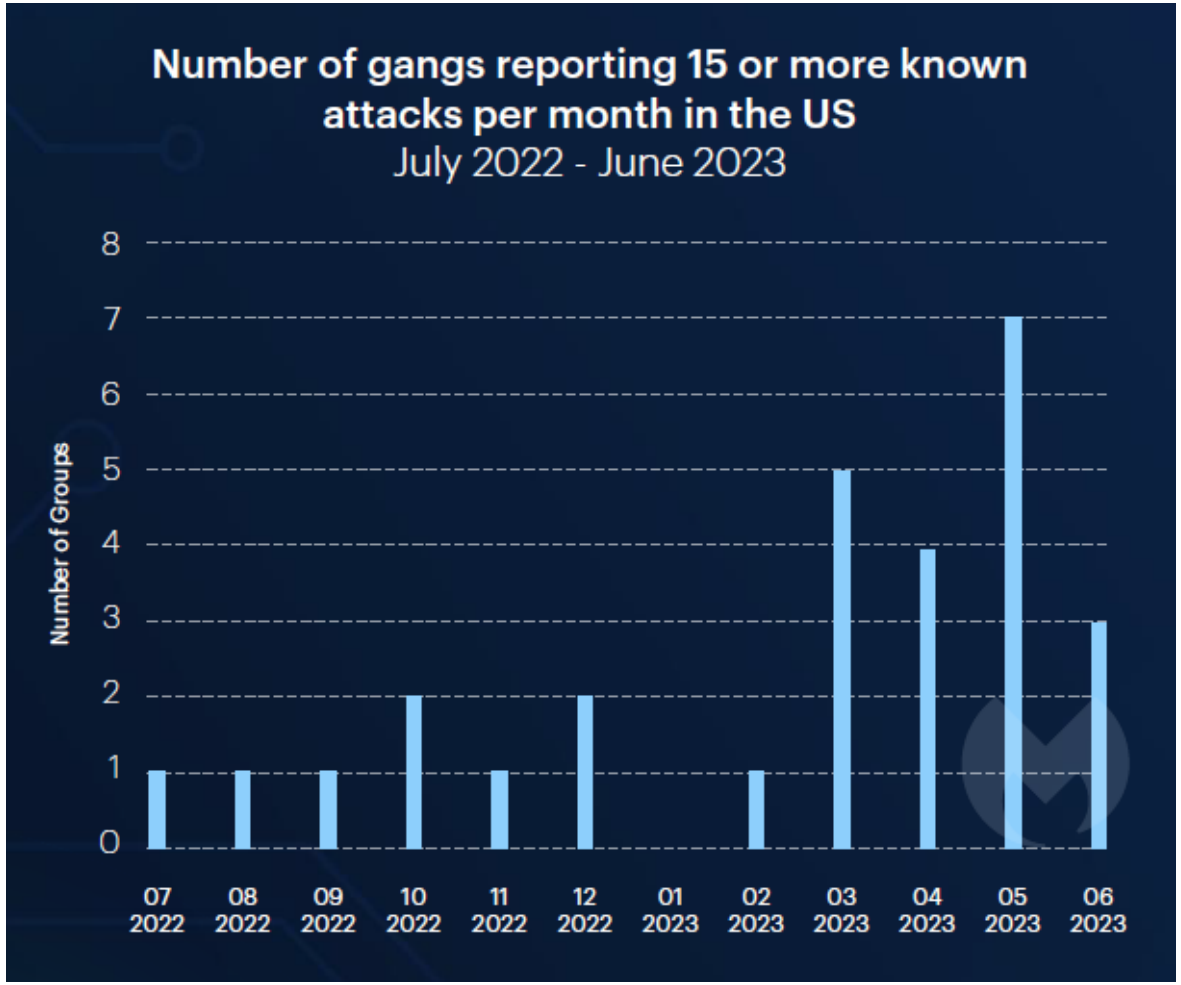


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Ransomware continues to be a Threat

- Ransomware Gangs

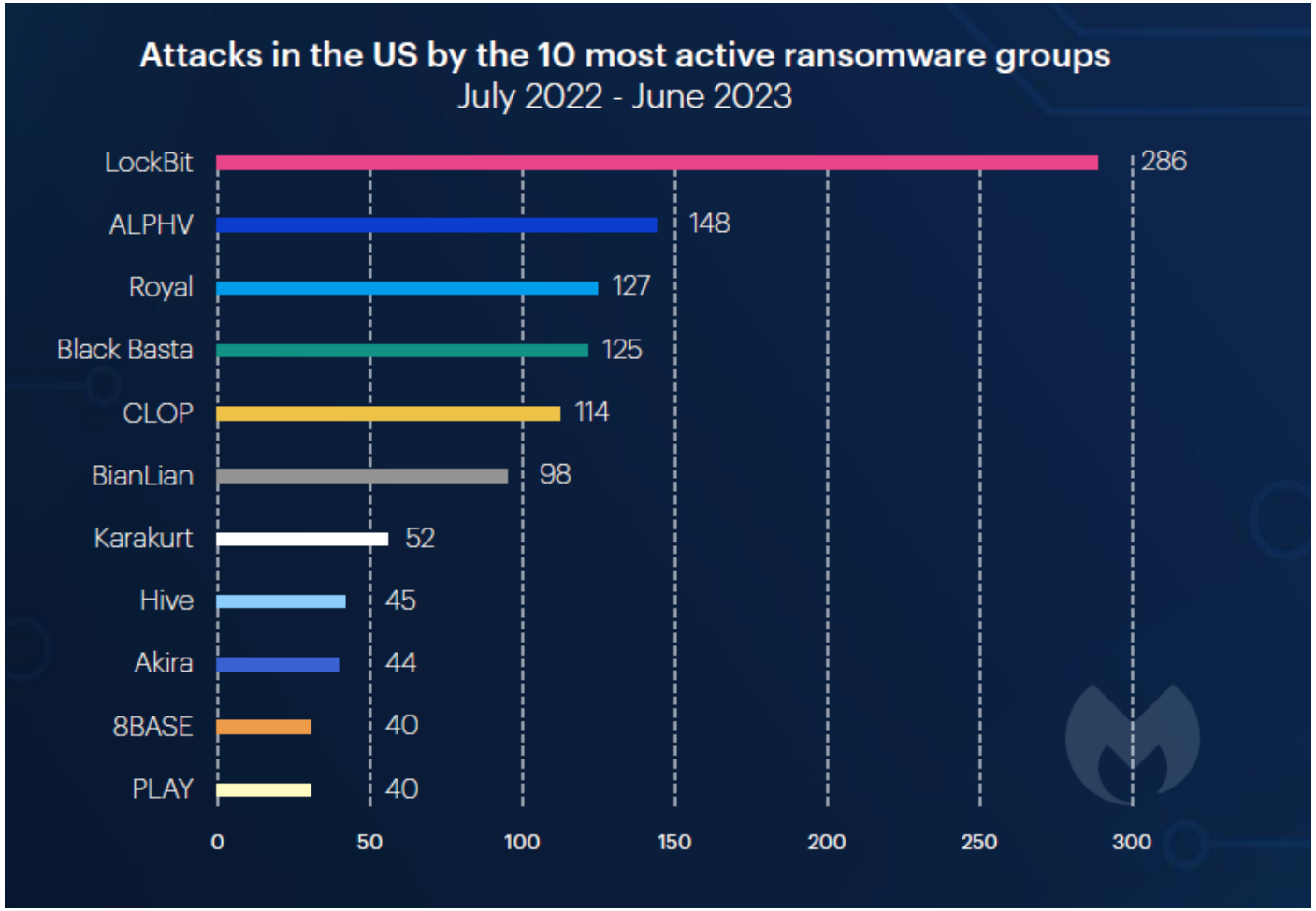


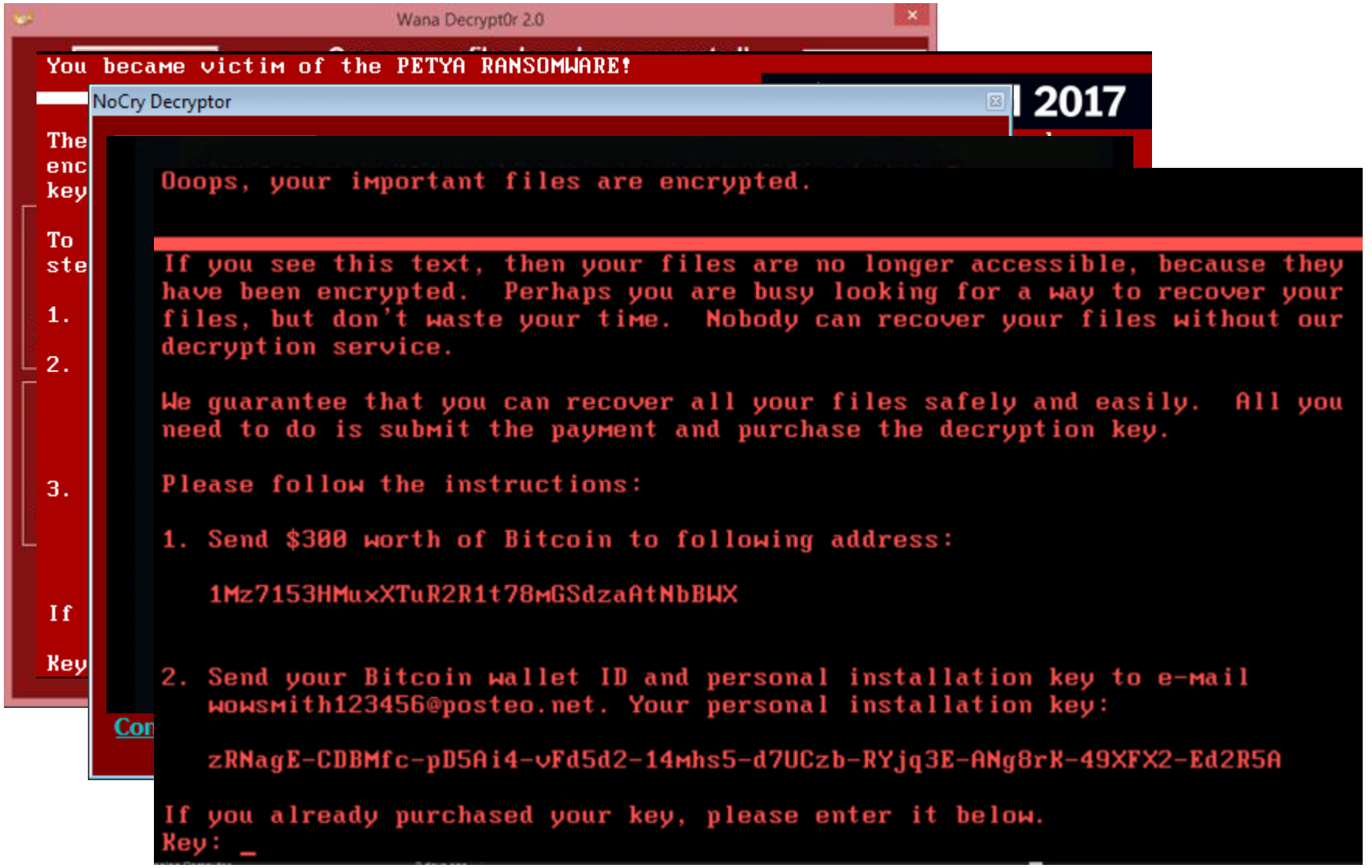
Image Credit: *Malwarebytes – State of Ransomware Report 2023*

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Circa 2016

- Top 10 Worst Ransomware Attacks

1. NotPetya
2. WannaCry
3. Locky
4. CrySis
5. Nemucod
6. Jaff
7. Spora
8. Cerber
9. Cryptomix
10. Jigsaw



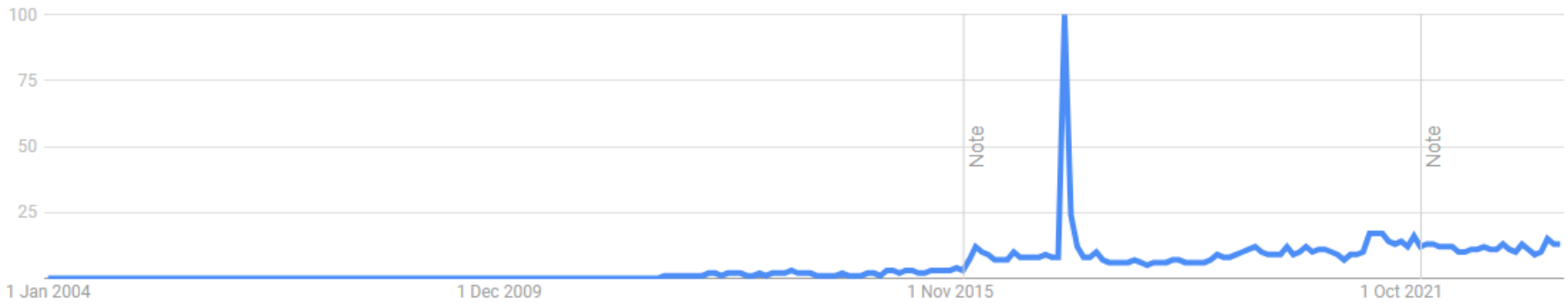
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ransomware
Search term

+ Compare

Worldwide | 01/01/2004 - 06/11/2023 | All categories | Web Search

Interest over time



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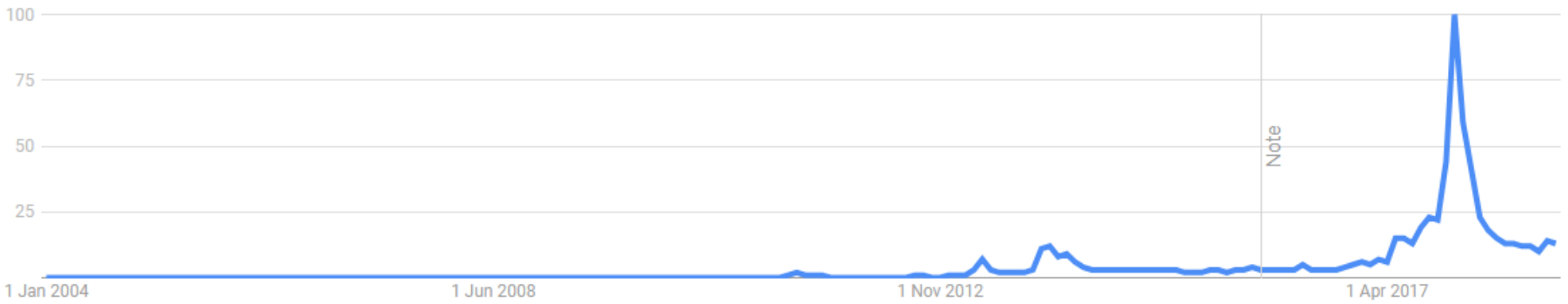
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Bitcoin
Search term

+ Compare

Worldwide | 01/01/2004 - 31/12/2018 | All categories | Web Search

Interest over time



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ransomware
Search term

Bitcoin
Search term

+ Add comparison

Worldwide | 01/01/2004 - 31/12/2018 | All categories | Web Search

Interest over time



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Innovation

Devising an Anti-ransomware Solution

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Strategies



Prevention

Block Malware

Zero-day Ransomware?

Polymorphic Malware?



Avoidance

Cloud Storage

(e.g. Microsoft OneDrive)

Minimize impact → Restore to last known-good backup

Incur re-image cost & effort

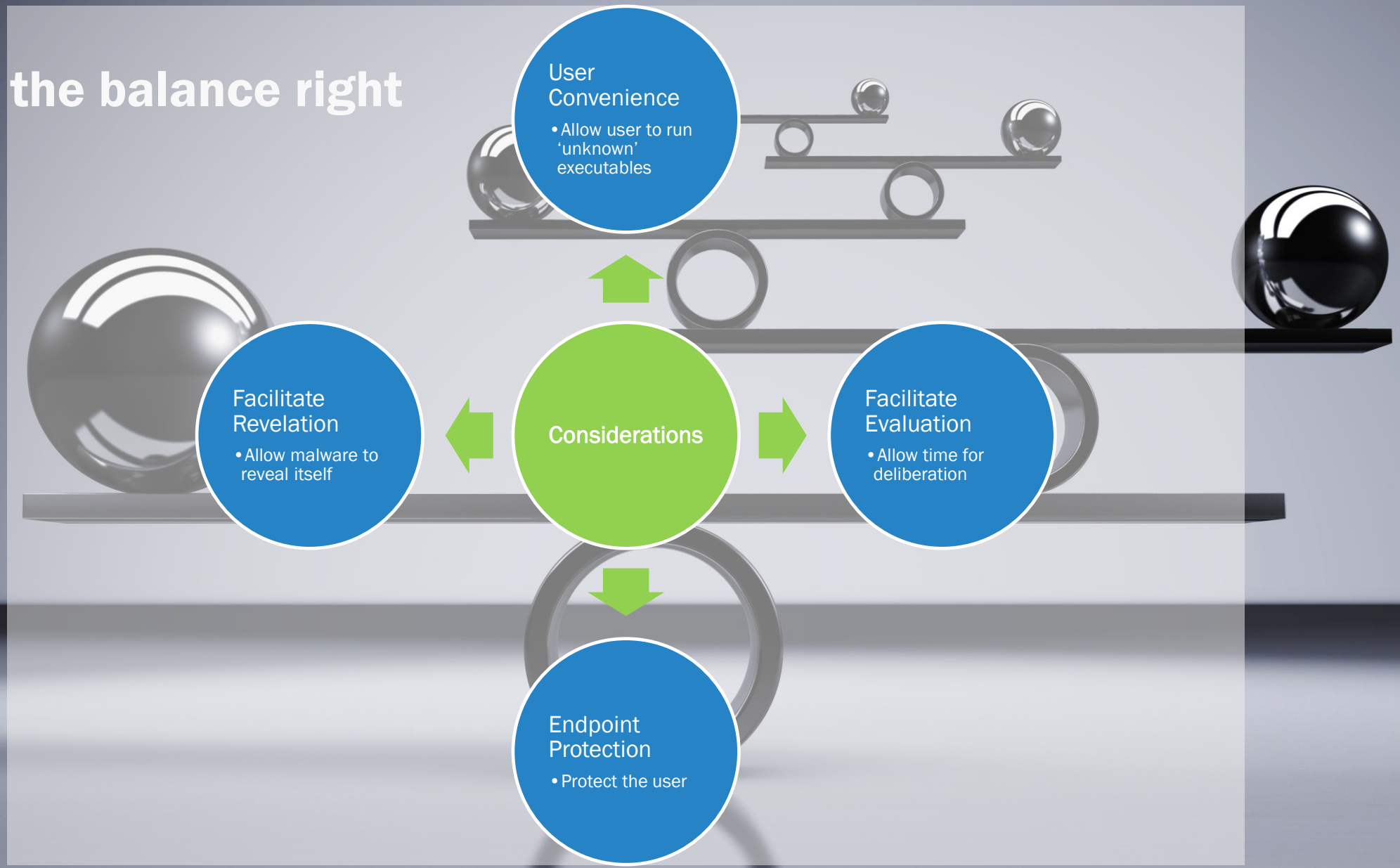


Remediate

Restore Endpoint to previous known-good state

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Getting the balance right



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REMEDIATION STRATEGIES



Confinement

Limit damage to a Virtual Image

Ditch VM on conviction

Roll forward if not convicted



Rollback Reversal

Perfect Undo

Rollback on conviction

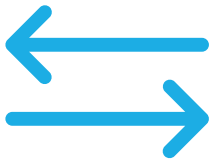
Do nothing if not convicted

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Remediation Strategies



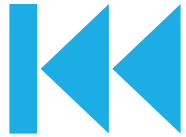
Optimistic v Pessimistic View



Roll forwards – or – Roll backwards



Most software != malware



Fewer instances where we need to take action → Roll backwards

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Rollback Remediation

What is it? How does it work?

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Rollback Remediation



Before Conviction

Monitor

Create Backups



Post Conviction

Kill Process Tree

Restore

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Rollback Remediation



Correction Technology

Does not detect

Does not protect



Passively Monitors

Performs Low-level Monitoring

Maintains ‘Process Genealogy’ Tree



Active Monitoring Triggered

Records detailed activity

Maintains copies of File, Directory and Registry changes.

Monitors child processes, proxies or any other surrogates.



Remediates

Terminates Process Tree

Uses Logs to restore all File, Directory & Registry changes & perform rollback.

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Remediation Details



File & Directory

Files & Directories restored to original state.

Encrypted files removed.

‘Dropped files’ removed.



Windows Registry

RegKeys & Values restored to original state.

Created RegKeys & Values removed.



File Attributes

Preserve Hidden/Archive file attributes.



Security Descriptors

Preserve file security metadata.

Restore File/Directory ownership.



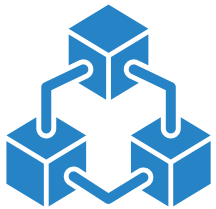
RunKeys & Startup Directories

Clean autostart registrations.

Clean known (and not-so-well known) directory & registry locations.

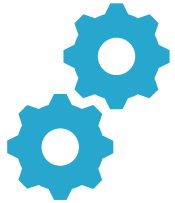
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Detecting Lateral Movement*



Windows Service Control Manager

Registration of background Windows Services (Daemons).



Windows Task Scheduler

Remove tasks scheduled for future execution.



Windows BITS

Background Intelligent Transfer Service

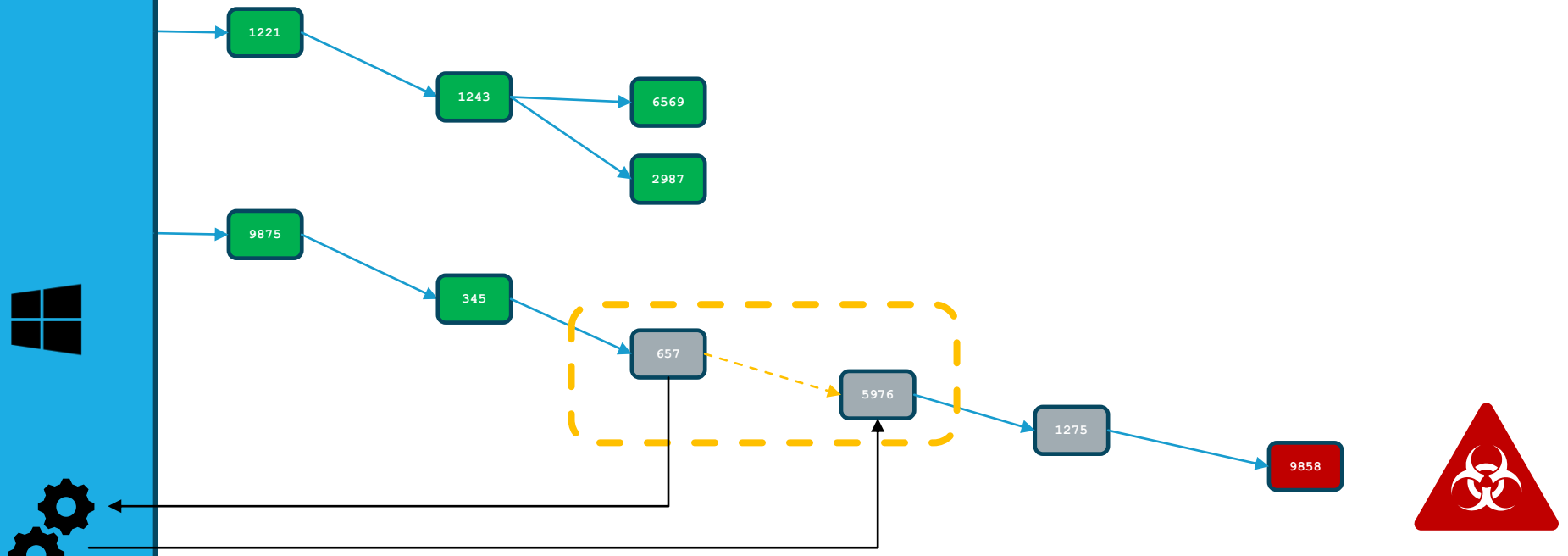
Facilitate intelligent file download

Download failure triggers registered handler.

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*Endpoint Lateral Movement (or Micro Lateral Movement)

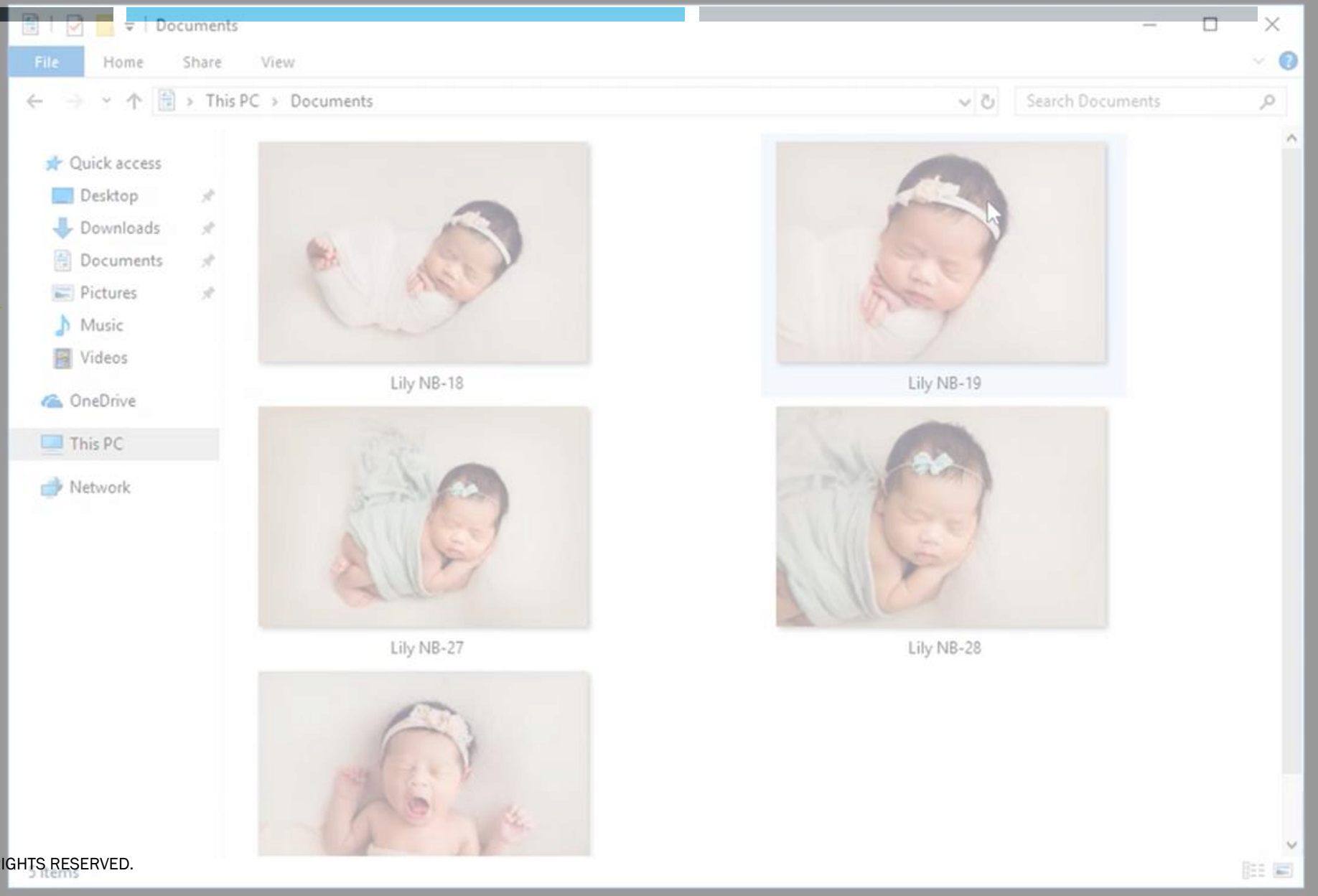
Detecting Lateral Movement



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DEMO

https://www.youtube.com/watch?v=OwJSZT2U4kM



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Conclusion

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Summary

- What & Why of Endpoint Security
- Threat Actors, Markets & Challenges
- Better Understanding of how how AV works
- Understand the Ransomware Threat & Explored Anti-ransomware



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Key Takeaways

- Importance of multi-faceted defence of the Endpoint.
- Extensive Backend Infrastructure
- Threat Landscape is constantly evolving.
- Endpoint Security more important today than ever.
- No one vendor has all the solutions.



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Thank you

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Engineering

Building Solutions for GA

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Timeline



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ENGINEERING PRACTICES



Agile/Scrum

Daily Standup

2-week Sprint

6-week PSI (Potentially Shippable Increment)

End-of-Sprint Demo

Sprint Planning

Proxy Product Owner



Test Driven Development

Pair Engineers with SDET

Automated Functional Tests

Integrated into build

1000+ Test Cases



Continuous Build

Daily/Weekly/On-demand (PR)

Integrated Tests



Virus Labs

Test against specific Malware

Remediate Top-10 Ransomware

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Performance Engineering

Ensuring ‘cost’ is within bounds

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Performance Engineering



Endpoint is ‘highly’ sensitive to resource utilization

Active monitoring costs

e.g. File copies



EPP should be in the background

Operation imperceptible to the user



Public Tests rule the day

AV-TEST

AV-Comparisons

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Performance Engineering



Real-world Scenarios

Install Browser

Install LibreOffice

Uncompress operation



Quantitative Loads

Determine if performance is linear or potentially exponential

Variables = No. of Files, Size/Type of Files, Directory depth/breath

Scientific Approach – i.e. modify 1 variable at a time.



User Profile

Define typical profiles for light, medium & heavy users

e.g. Browsing & Streaming, MS Office user, Software Engineer



Windows ADK

Internal Standard

Common Use-cases

(e.g. browser load, application launch, boot-up, restart, etc.)

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Performance Engineering



Manage Performance Metrics

Capture & Store performance data

Graphically display results



Performance Trends

Track performance data over time.

Determine general improvements

Capture anomalies/outliers



CI/CT

Continuous Testing

Integrated into build

Hourly/daily/weekly builds

Show trend

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ransomware
Search term

+ Compare

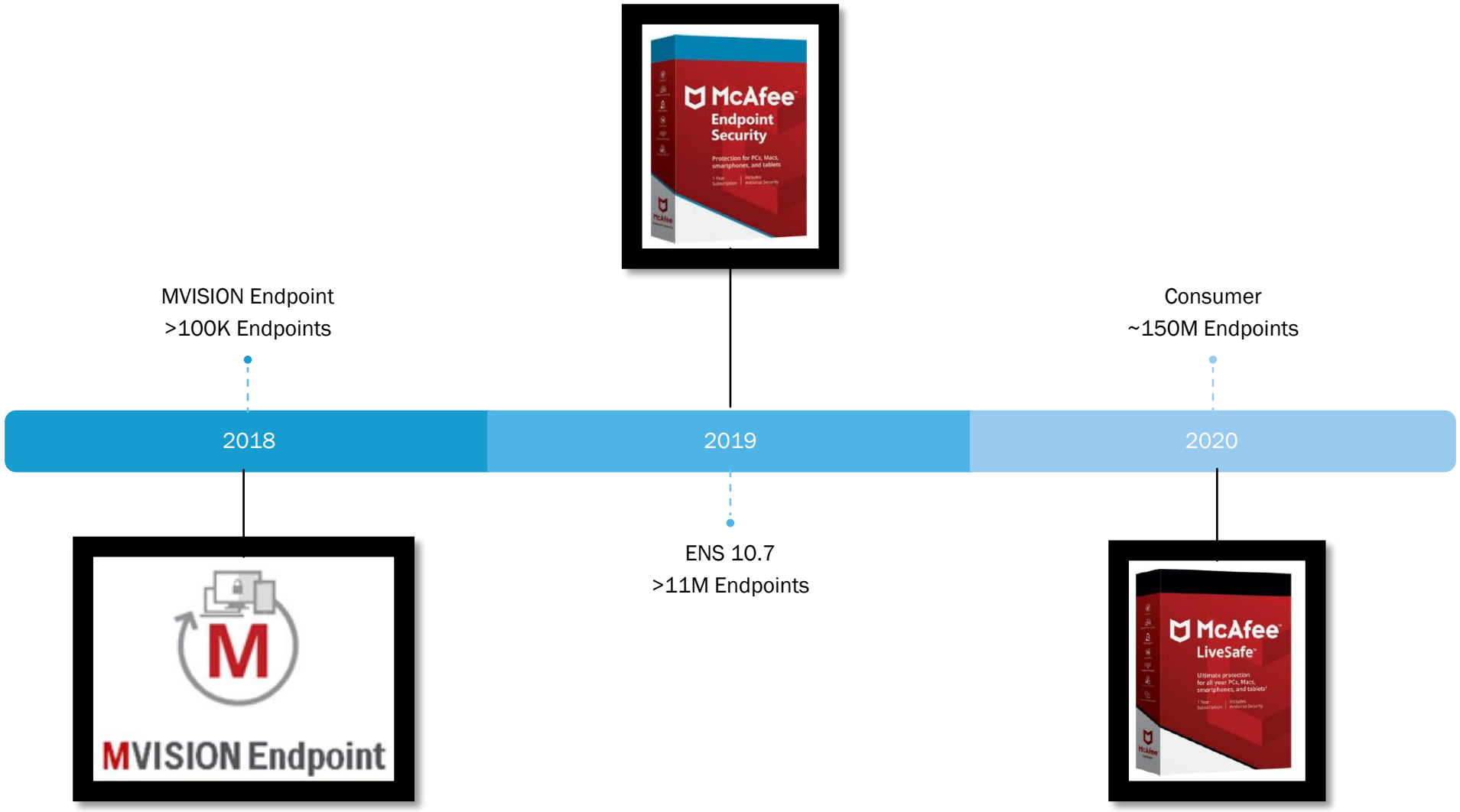
Worldwide | 01/01/2004 - 31/12/2018 | All categories | Web Search

Interest over time



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Integration Timeline



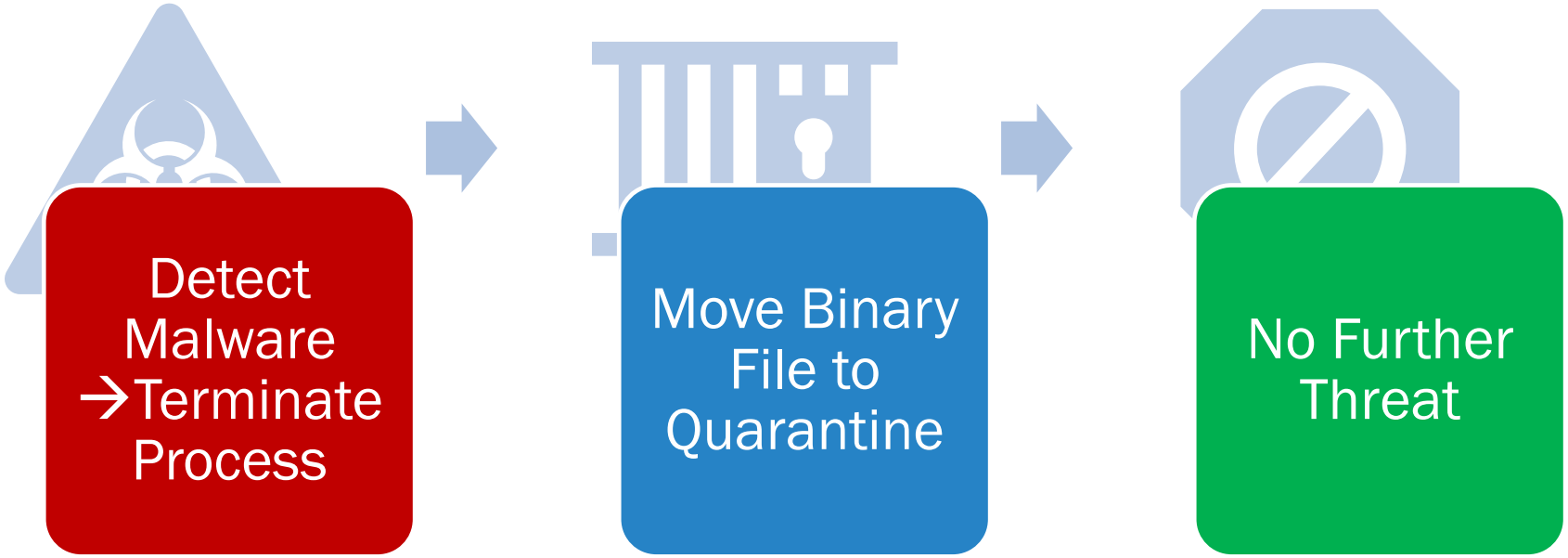
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Fileless Remediation

Breaking the Execution Chain

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Normal AV



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Systems

System Tree

Threat Event Log: Details — Previous | Next

Threat Handled:	True
Analyzer Detection Method:	MVISION Endpoint

Events received from managed systems

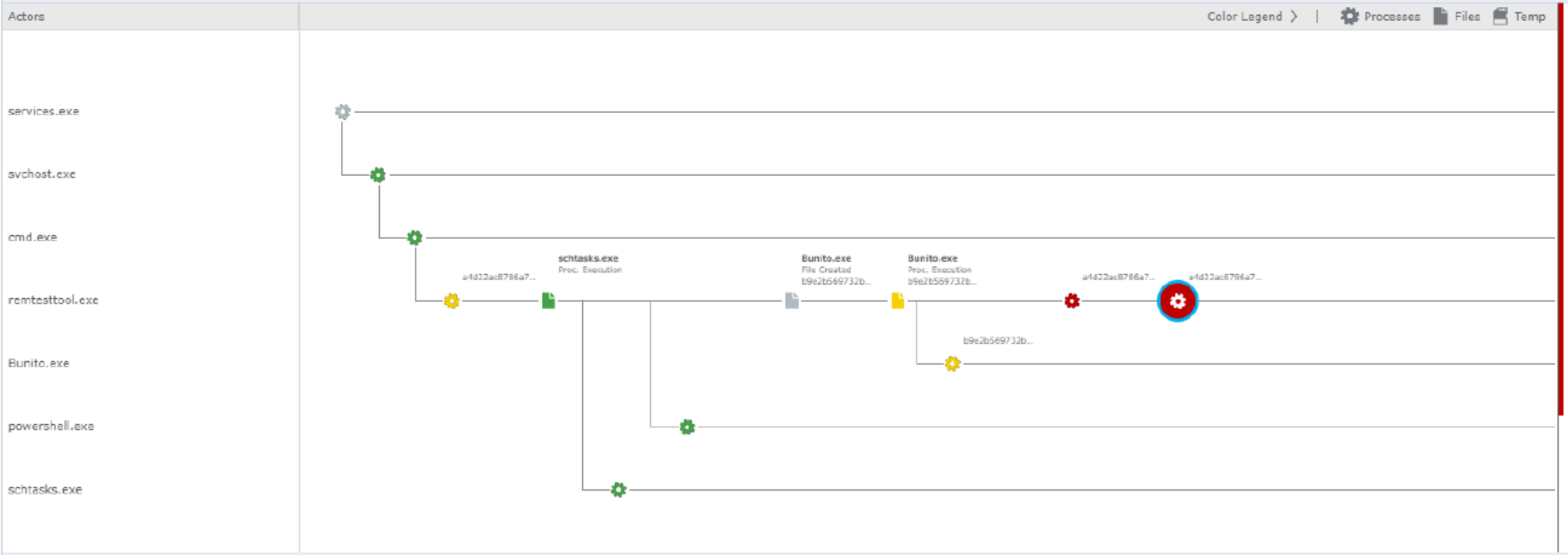
Event Description:	Infected file was moved to quarantine area.

Endpoint Security

Adaptive Threat Protection Events

Web Control Events

Story Graph (Trace Summary)



Event Details

Process terminated

Target Name	C:\Users\Ted\Desktop\deep_remediation_sample\remtesttool.exe
Reputation	Known Malicious
Reputation Score	1
PID	2964:132158147741768607

Action Taken
Threat has been quarantined

SHA-256
a4d22ac8786a75350f8ce8cc9f0f493fae227a6ff1e68945b23735f413785f49

MD5
4bf25e06e9e6c86e797da9a0cbc72fbb

Command Line Parameter
.\remtesttool.exe -sb -ib -wb -r -w1 18000 -db -rb -w2 180000

MVISION Endpoint

Target MD5 Hash:	4bf25e06e9e6c86e797da9a0cbc72fbb
Target SHA-256 Hash:	a4d22ac8786a75350f8ce8cc9f0f493fae227a6ff1e68945b23735f413785f49

Related Items

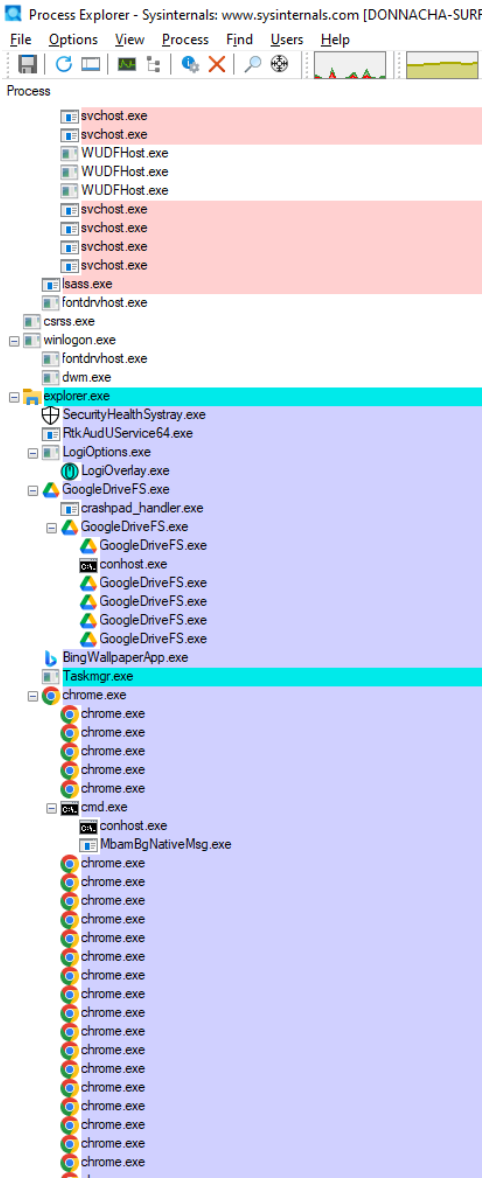
Go to related Systems

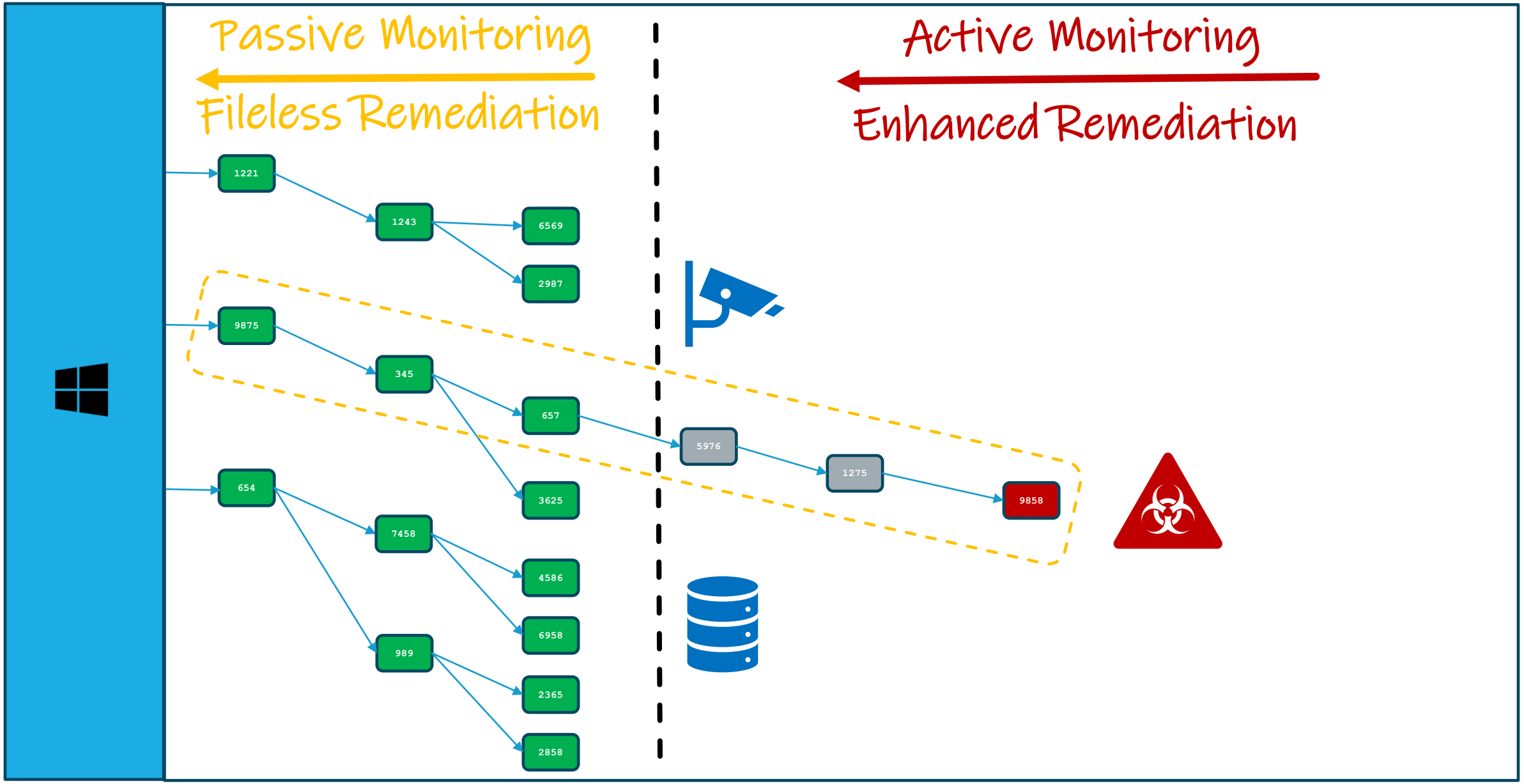
Actions | Show Source Systems | Show Target Systems

Back | Close

Process Tree Hierarchy

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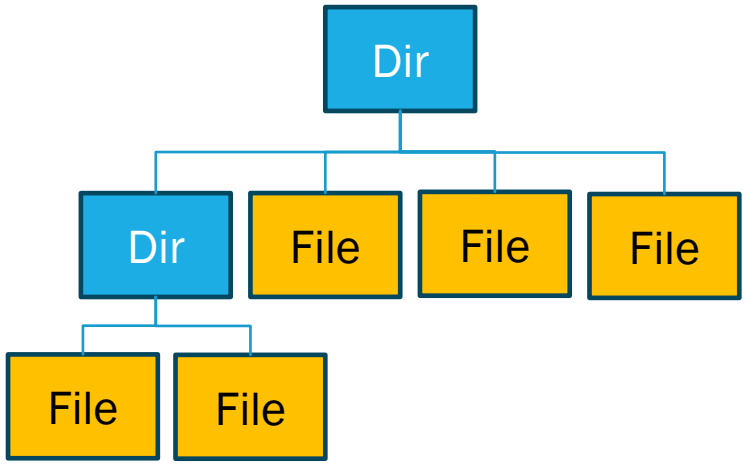


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How Ransomware Works

Iterate over the File System



*Focus on typical document locations
*Target office docs

Pseudo Code

Open File System

- Open Directory
- Read File
- Write Encrypted File
- Delete Original
- Loop

Loop

What's the Problem?

Zero-day!

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